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1 Mason and Grijalva

Table 1: MRQ6 recoded audit table for Mason and Grijalva (2019). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
MasonGrijalva2019	1	2	1	The review is infrastructure-aware but only lightly reproducibility-critical. Reproducibility auditing receives score 1 because the paper does not systematically audit open code, benchmark comparability, dataset openness, artifact availability, or reporting standards. Infrastructure auditing receives score 2 because the paper discusses the wider technical stack needed for autonomous building energy management, including sensors, actuators, communication infrastructure, IoT, smart meters, batteries, smart-grid integration, simulator design, electricity prices, weather, PV variability, and occupancy-related state information. Blind-spot visibility receives score 1 because simulator design and transfer learning are identified as concerns for scaling RL beyond single-building controller training, but the paper does not develop these issues as a strong reproducibility or shared-infrastructure agenda.

Table 2: MRQ6 infrastructure-coverage scores for Mason and Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Mason and Grijalva (2019)	2	0	0	0	2	1

Table 3: MRQ6 critical-audit scores for Mason and Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Mason and Grijalva (2019)	0	0	1	0	0

Table 4: MRQ6 recurring infrastructure-gap scores for Mason and Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Mason and Grijalva (2019)	0	0	0	1	0

Table 5: MRQ6 summary table for Mason and Grijalva (2019), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Mason and Grijalva (2019)	Infrastructure-aware but only lightly reproducibility-critical	The survey’s main infrastructure concern is that reinforcement-learning-based building control depends on a wider technical stack, especially simulator design, smart-metering, IoT connectivity, and grid-linked external information such as prices and weather.	The survey contributes to MRQ6 mainly by showing that reinforcement learning for building energy management rests on nontrivial technical infrastructure and external-signal assumptions, but it does not yet develop a strong audit of reproducibility, benchmark comparability, open artifacts, or dataset standardization.

Table 6: One-line MRQ6 answer for Mason and Grijalva (2019), after recoding.

survey_id	MRQ6 answer
MasonGrijalva2019	Mason and Grijalva (2019) address MRQ6 mainly by showing that reinforcement-learning-based building energy management depends on simulators, IoT-connected devices, smart meters, and external signals such as weather and pricing, while only lightly auditing reproducibility itself and giving little direct attention to open code, shared benchmarks, or dataset standardization.

2 Wang and Hong

Table 7: MRQ6 recoded audit table for Wang and Hong (2020). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
WangHong2020	3	3	3	The review gives a strong infrastructure and reproducibility critique. Reproducibility auditing receives score 3 because the paper structurally audits training-data sources, environmental settings, simulator dependence, real-building implementation, comparison baselines, and cross-study comparability. Infrastructure auditing receives score 3 because the paper discusses simulators, actual buildings, open-source control testbeds, Open Building Controls, CityLearn, external signals, and shared datasets as core infrastructure for RL-based building control. Blind-spot visibility receives score 3 because the paper explicitly argues that fair comparison is difficult without common, credible benchmarks and calls for open-source and well-recognized control testbeds and datasets.

Table 8: MRQ6 infrastructure-coverage scores for Wang and Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Wang and Hong (2020)	3	1	3	1	2	2

Table 9: MRQ6 critical-audit scores for Wang and Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Wang and Hong (2020)	3	1	2	2	3

Table 10: MRQ6 recurring infrastructure-gap scores for Wang and Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Wang and Hong (2020)	0	3	2	2	1

Table 11: MRQ6 summary table for Wang and Hong (2020), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Wang and Hong (2020)	Benchmark- and infrastructure-aware, with a clear reproducibility critique	The survey’s dominant infrastructure concern is the lack of common and credible benchmarking infrastructure, given the heavy dependence on simulators, learning-based building-control research is not only simulation-heavy, but also inconsistent comparison baselines, weak real-building adoption, and the need for open-source testbeds and datasets.	The survey contributes strongly to MRQ6 by showing that reinforcement-learning-based building-control research is not only simulation-heavy, but also difficult to compare fairly across studies without shared evaluation environments, better external-signal handling, and more standardized benchmarking practices.

Table 12: One-line MRQ6 answer for Wang and Hong (2020), after recoding.

survey_id	MRQ6 answer
WangHong2020	Wang and Hong (2020) answer MRQ6 by explicitly showing that reinforcement-learning-based building-control research depends heavily on simulators, lacks shared and credible benchmarks for fair comparison, underuses important external signals such as forecasts, and therefore needs open-source datasets and standardized testbeds to improve reproducibility and infrastructure quality.

3 Han et al.

Table 13: MRQ6 recoded audit table for Han et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
HanEtAl2021	1	2	2	The review is data- and infrastructure-aware, but only lightly reproducibility-critical. Reproducibility auditing receives score 1 because the paper does not systematically audit open code, artifact availability, shared benchmarks, or reproducible experimental packages. Infrastructure auditing receives score 2 because the paper discusses high-quality data, historical occupant behavior information, repeated occupant feedback, frequent state and reward signals, transfer learning, function approximation, and collaboration for sharing data and research infrastructures. Blind-spot visibility receives score 2 because the paper explicitly identifies weak validation standards, data limitations, context dependence, and the need for stronger collaboration as constraints for occupant-behavior-oriented RL.

Table 14: MRQ6 infrastructure-coverage scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Han et al. (2021)	1	3	0	0	1	2

Table 15: MRQ6 critical-audit scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Han et al. (2021)	0	0	0	2	1

Table 16: MRQ6 recurring infrastructure-gap scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Han et al. (2021)	0	0	1	0	2

Table 17: MRQ6 summary table for Han et al. (2021), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Han et al. (2021)	Data- and infrastructure-aware, but only lightly reproducibility-critical	The survey’s dominant infrastructure concern is the weak standardization of validation practice and the heavy dependence of occupant-centered reinforcement learning on sufficient high-quality data, repeated feedback signals, and collaboration around shared research infrastructure.	The survey contributes to MRQ6 mainly by showing that occupant-behavior reinforcement learning still depends on fragile data and validation infrastructures, but it contributes less strongly to benchmark, code, and artifact reproducibility than the more infrastructure-auditing surveys.

Table 18: One-line MRQ6 answer for Han et al. (2021), after recoding.

survey_id	MRQ6 answer
HanEtAl2021	Han et al. (2021) answer MRQ6 mainly by showing that reinforcement learning for occupant behavior relies on weakly standardized validation practice, high-quality data, repeated feedback signals, and stronger collaboration around shared knowledge and infrastructures, while giving only limited attention to shared benchmarks, open artifacts, or code reproducibility.

4 Yu et al.

Table 19: MRQ6 recoded audit table for Yu et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
YuEtAl2021	1	2	1	The review is training- and modeling-infrastructure aware, but only lightly reproducibility-critical. Reproducibility auditing receives score 1 because the paper does not strongly audit open code, artifact availability, shared benchmark tasks, or reusable experimental packages. Infrastructure auditing receives score 2 because the paper discusses the dependence of DRL-based smart building energy management on simulations, calibrated environment models, historical operational data, IoT-collected data, cloud computing, big-data analytics, and data-efficient training pipelines. Blind-spot visibility receives score 1 because costly data collection and inefficient direct interaction with real buildings are recognized as barriers, but the paper does not turn these into a strong shared-benchmark or open-artifact agenda.

Table 20: MRQ6 infrastructure-coverage scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Yu et al. (2021)	3	2	1	0	2	3

Table 21: MRQ6 critical-audit scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Yu et al. (2021)	1	0	1	2	1

Table 22: MRQ6 recurring infrastructure-gap scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Yu et al. (2021)	0	1	1	1	1

Table 23: MRQ6 summary table for Yu et al. (2021), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Yu et al. (2021)	Training- and modeling-infrastructure aware, but only moderately reproducibility-critical	The survey’s dominant infrastructure concern is the dependence of DRL-based smart building energy management on environment models, calibrated simulation frameworks, historical or IoT-collected data, and data-efficient training pipelines.	The survey contributes to MRQ6 mainly by making the technical infrastructure of DRL-based smart building energy management explicit and by showing that practical applicability depends heavily on data efficiency and environment-model support, but it contributes less strongly to open-code, artifact, and shared-benchmark reproducibility critique.

Table 24: One-line MRQ6 answer for Yu et al. (2021), after recoding.

survey_id	MRQ6 answer
YuEtAl2021	Yu et al. (2021) answer MRQ6 mainly by showing that DRL for smart building energy management relies on substantial modeling and data infrastructure, including simulations, calibrated environment models, and IoT-collected data, while practical scalability is limited by data inefficiency and costly interaction with real buildings rather than by an explicitly developed shared-benchmark or open-artifact culture.

5 Sierla et al.

Table 25: MRQ6 recoded audit table for Sierla et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
SierlaEtAl2022	3	3	3	The review is strongly comparability- and reporting-oriented. Reproducibility auditing receives score 3 because the paper structurally evaluates how action formulation, abstraction level, control targets, and reporting sufficiency affect the comparability and reproducibility of RL-for-HVAC studies. Infrastructure auditing receives score 3 because the review distinguishes physical HVAC systems from virtual environments, including physics-based building simulators and data-driven models, and assesses how these choices affect deployability and interpretation. Blind-spot visibility receives score 3 because unclear actuator or setpoint mappings, excessive abstraction, vague simulator capabilities, and weak links between indoor-environment variables and HVAC action spaces are treated as structural blind spots.

Table 26: MRQ6 infrastructure-coverage scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Sierla et al. (2022)	3	1	1	0	2	3

Table 27: MRQ6 critical-audit scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Sierla et al. (2022)	3	0	3	1	3

Table 28: MRQ6 recurring infrastructure-gap scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Sierla et al. (2022)	0	2	2	3	1

Table 29: MRQ6 summary table for Sierla et al. (2022), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Sierla et al. (2022)	Comparability- and reporting-oriented, with a strong environment-realism critique	The survey’s dominant infrastructure concern is that studies differ too much in abstraction level, action-space design, simulator realism, virtual-versus-physical environment choice, and reporting clarity for results to be compared cleanly or interfaced confidently with real HVAC systems.	The survey contributes strongly to MRQ6 by showing that reproducibility problems in RL-for-HVAC are not only about open artifacts, but also about whether environments, action formulations, abstraction levels, and reporting detail are specified clearly enough to support comparison and later deployment.

Table 30: One-line MRQ6 answer for Sierla et al. (2022), after recoding.

survey_id	MRQ6 answer
SierlaEtAl2022	Sierla et al. (2022) answer MRQ6 mainly by showing that RL-for-HVAC studies are often hard to compare and reproduce at the control-design level because simulator realism, action formulations, abstraction levels, external assumptions, and reporting detail vary substantially across the literature, even when reward-focused results look promising.

6 Fu et al.

Table 31: MRQ6 recoded audit table for Fu et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
FuEtAl2022	1	2	1	The review is broadly infrastructure-aware but only lightly reproducibility-critical. Reproducibility auditing receives score 1 because the paper does not systematically evaluate open code, shared benchmark tasks, artifact availability, or dataset openness. Infrastructure auditing receives score 2 because the paper discusses simulation platforms as a practical response to the risks and costs of real-building training, including examples such as BCVTB with EnergyPlus and CitySim with TensorFlow. Blind-spot visibility receives score 1 because the paper recognizes simulator dependence, training time, environment modeling, and real-building training risk, but does not develop these into a strong reproducibility or shared-infrastructure critique.

Table 32: MRQ6 infrastructure-coverage scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Fu et al. (2022)	2	0	0	0	1	2

Table 33: MRQ6 critical-audit scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Fu et al. (2022)	0	0	1	0	0

Table 34: MRQ6 recurring infrastructure-gap scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Fu et al. (2022)	0	0	1	1	0

Table 35: MRQ6 summary table for Fu et al. (2022), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Fu et al. (2022)	Broadly infrastructure-aware, but only lightly reproducibility-critical	The survey’s dominant infrastructure concern is that reinforcement learning for building energy efficiency depends heavily on simulation platforms because real-building training is risky, costly, and time-consuming.	The survey contributes to MRQ6 mainly by showing that simulator dependence and training burden are central practical bottlenecks, but it remains weak on open-artifact availability, shared benchmarks, and dataset-centered reproducibility critique.

Table 36: One-line MRQ6 answer for Fu et al. (2022), after recoding.

survey_id	MRQ6 answer
FuEtAl2022	Fu et al. (2022) answer MRQ6 mainly by showing that reinforcement-learning-based building-energy-efficiency control still relies strongly on simulation environments because direct real-building training is risky and costly, while giving only limited attention to shared datasets, benchmark infrastructure, open code, or artifact reproducibility.

7 Shaqour and Hagishima

Table 37: MRQ6 recoded audit table for Shaqour and Hagishima (2022). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
ShaqourHagishima2022	3	3	3	The review is strongly infrastructure- and reproducibility-aware. Reproducibility auditing receives score 3 because the paper includes explicit discussion of reproducibility, open-sourcing code, standardized open datasets, and the need to improve comparison and reuse. Infrastructure auditing receives score 3 because the paper discusses CityLearn as an open-source environment intended to support standardized comparison of reinforcement-learning algorithms for building energy management. Blind-spot visibility receives score 3 because the paper links data hunger, limited real-world validation, lack of openness, and insufficient standardization to weak comparability and reproducibility in DRL-based BEMS research.

Table 38: MRQ6 infrastructure-coverage scores for Shaqour and Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Shaqour and Hagishima (2022)	3	3	3	3	2	3

Table 39: MRQ6 critical-audit scores for Shaqour and Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Shaqour and Hagishima (2022)	3	3	1	3	3

Table 40: MRQ6 recurring infrastructure-gap scores for Shaqour and Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Shaqour and Hagishima (2022)	3	3	1	1	3

Table 41: MRQ6 summary table for Shaqour and Hagishima (2022), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Shaqour and Hagishima (2022)	Strongly infrastructure- and reproducibility-aware, especially for an early survey	The survey’s dominant infrastructure concern is that DRL-based BEMS remains constrained by large data requirements, scarce real validation, cross-building-type mapping with an explicit call for open-source envi- and insufficiently standardized open benchmarking and sharing prac- tices.	The survey contributes strongly to MRQ6 by combining systematic ronments, open code, and standardized datasets, showing that infras- tructure weakness in the field is not only about scarce deployment but also about limited comparability and reuse.

Table 42: One-line MRQ6 answer for Shaqour and Hagishima (2022), after recoding.

survey_id	MRQ6 answer
ShaqourHagishima2022	Shaqour and Hagishima (2022) answer MRQ6 by showing that DRL-based BEMS research is still constrained by data hunger and weak real-world validation, while also arguing that stronger reproducibility and comparability require open-source environments such as CityLearn, open code, and more standardized open datasets.

8 Zhang et al.

Table 43: MRQ6 recoded audit table for Zhang et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
ZhangEtAl2022	1	2	2	The review is control- and modeling-infrastructure aware, but only lightly reproducibility-critical. Reproducibility auditing receives score 1 because the paper does not systematically audit open code, shared benchmark tasks, artifact availability, or reproducible experimental packages. Infrastructure auditing receives score 2 because the paper discusses data-driven MPC and RL for BEMS, historical operational data, MPC-generated training data, accurate simulators, learned environment models, forecasts, future disturbances, occupancy, human interaction, and external uncertainty. Blind-spot visibility receives score 2 because the paper identifies data efficiency, safety, robust adaptability, simulator cost, and learning from historical data as practical limitations, but does not develop a strong shared-benchmark or open-artifact reproducibility agenda.

Table 44: MRQ6 infrastructure-coverage scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Zhang et al. (2022)	2	2	0	0	2	2

Table 45: MRQ6 critical-audit scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Zhang et al. (2022)	0	0	1	2	0

Table 46: MRQ6 recurring infrastructure-gap scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Zhang et al. (2022)	0	0	1	1	1

Table 47: MRQ6 summary table for Zhang et al. (2022), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Zhang et al. (2022)	Control- and modeling-infrastructure aware, but only lightly reproducibility-critical	The survey’s dominant infrastructure concern is that practical BEMS control depends on data sources, forecast information, simulators, learned environment models, and historically collected operational data, while data efficiency and safety remain unresolved.	The survey contributes to MRQ6 mainly by making the data, model, simulator, and forecast infrastructure behind MPC- and RL-based BEMS more explicit, but it remains weaker on shared benchmarks, open artifacts, and direct reproducibility critique than the more infrastructure-auditing surveys.

Table 48: One-line MRQ6 answer for Zhang et al. (2022), after recoding.

survey_id	MRQ6 answer
ZhangEtAl2022	Zhang et al. (2022) answer MRQ6 mainly by showing that BEMS control research depends on nontrivial data, model, simulator, and forecast infrastructure, while practical RL remains limited by data efficiency, safety, robust adaptability, and the cost of accurate environment modeling rather than by an explicitly developed benchmark or open-artifact culture.

9 Al-Ani and Das

Table 49: MRQ6 recoded audit table for Al-Ani and Das (2022). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
AlAniDas2022	1	2	1	The review is broadly infrastructure-aware but weakly reproducibility-critical. Reproducibility auditing receives score 1 because the paper does not systematically audit open code, shared datasets, benchmark environments, artifact availability, or standardization needs across the literature. Infrastructure auditing receives score 2 because the paper discusses HEMS enabling technologies and infrastructure, including communication protocols, IoT-enabled devices, wireless communication, HAN/WAN structures, interoperability, scalability, simulation/model-based training, and limited actual HEMS deployment. Blind-spot visibility receives score 1 because the paper signals that RL in HEMS remains early-stage and largely simulation-based, but does not develop this into a strong reproducibility, open-artifact, or shared-benchmark critique.

Table 50: MRQ6 infrastructure-coverage scores for Al-Ani and Das (2022), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Al-Ani and Das (2022)	2	0	0	0	1	2

Table 51: MRQ6 critical-audit scores for Al-Ani and Das (2022), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Al-Ani and Das (2022)	0	0	0	1	0

Table 52: MRQ6 recurring infrastructure-gap scores for Al-Ani and Das (2022), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Al-Ani and Das (2022)	0	0	0	0	1

Table 53: MRQ6 summary table for Al-Ani and Das (2022), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Al-Ani and Das (2022)	Broadly infrastructure-aware, but weakly reproducibility-critical	The survey’s dominant infrastructure concern is that HEMS reinforcement learning remains early-stage and depends on enabling communication, IoT, and home-network infrastructure, while still relying largely on model-based or simulation-based training rather than mature real-world experimentation.	The survey contributes to MRQ6 mainly by signaling the immaturity and simulation dependence of the HEMS reinforcement-learning literature, but it remains weak on datasets, shared benchmarks, open artifacts, and direct reproducibility critique.

Table 54: One-line MRQ6 answer for Al-Ani and Das (2022), after recoding.

survey_id	MRQ6 answer
AlAniDas2022	Al-Ani and Das (2022) answer MRQ6 mainly by showing that reinforcement learning in HEMS is still early-stage and largely simulation-based, while depending on communication, IoT, and home-network infrastructure, and giving only limited attention to datasets, benchmark environments, shared artifacts, and reproducibility infrastructure needed for stronger cross-study comparability.

10 Weinberg et al.

Table 55: MRQ6 recoded audit table for Weinberg et al. (2023). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
WeinbergEtAl2023	2	3	3	The review is system- and infrastructure-aware, but only moderately reproducibility-critical. Reproducibility auditing receives score 2 because the paper does not audit open artifacts or common benchmark datasets, but it structurally discusses sample efficiency, slow data collection, transfer difficulty, and data-processing conditions that affect comparability and practical reproducibility. Infrastructure auditing receives score 3 because the paper explicitly frames practical RL for building energy systems around large-scale data collection, BMS sensing and actuation, scalable computation pipelines, cloud-based processing, communication between sensors and actuators, interoperability, and asynchronous multi-building data streams. Blind-spot visibility receives score 3 because sample-efficiency constraints, heterogeneous building dynamics, transfer difficulty, interoperability barriers, and scalable system design are structural computer-science challenges and future directions in the review.

Table 56: MRQ6 infrastructure-coverage scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Weinberg et al. (2023)	2	3	0	0	2	2

Table 57: MRQ6 critical-audit scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Weinberg et al. (2023)	0	0	1	3	3

Table 58: MRQ6 recurring infrastructure-gap scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Weinberg et al. (2023)	0	0	1	2	3

Table 59: MRQ6 summary table for Weinberg et al. (2023), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Weinberg et al. (2023)	System- and infrastructure-aware, but only moderately reproducibility-critical	The survey’s dominant infrastructure concern is that practical reinforcement learning for building energy systems depends on slow and costly data collection, transferable environment knowledge, scalable processing, communication infrastructure, interoperability, and asynchronous multi-building data streams.	The survey contributes to MRQ6 mainly by showing that reproducibility and comparability in reinforcement learning for building energy systems are constrained by sample efficiency, transfer difficulty, heterogeneous building dynamics, asynchronous data pipelines, and interoperability challenges, even though the paper is less focused on open artifacts or shared benchmark ecosystems.

Table 60: One-line MRQ6 answer for Weinberg et al. (2023), after recoding.

survey_id	MRQ6 answer
WeinbergEtAl2023	Weinberg et al. (2023) answer MRQ6 mainly by showing that reinforcement learning for building energy systems relies on demanding data-collection, transfer, communication, and computation infrastructure, and that practical progress is limited less by missing algorithm ideas than by sample-efficiency constraints, heterogeneous building dynamics, and the difficulty of scaling interoperable control pipelines.

11 Chatterjee and Khovalyg

Table 61: MRQ6 recoded audit table for Chatterjee and Khovalyg (2023). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
ChatterjeeKhovalyg2023	3	3	3	The review is strongly simulator-, reporting-, and benchmarking-aware. Reproducibility auditing receives score 3 because the paper uses explicit inclusion criteria, records building type, HVAC system, DRL algorithm, simulation framework, controlled parameters, and implementation status, and makes these reporting dimensions part of its synthesis. Infrastructure auditing receives score 3 because simulation software, building model scale, building type, virtual environments, and real-building implementation status are structurally visible in the review. Blind-spot visibility receives score 3 because the paper explicitly identifies the lack of a data-rich, open-source, interoperable virtual testbed as a barrier to cross-study validation and benchmarking.

Table 62: MRQ6 infrastructure-coverage scores for Chatterjee and Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Chatterjee and Khovalyg (2023)	3	2	3	1	2	3

Table 63: MRQ6 critical-audit scores for Chatterjee and Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Chatterjee and Khovalyg (2023)	3	1	3	2	3

Table 64: MRQ6 recurring infrastructure-gap scores for Chatterjee and Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Chatterjee and Khovalyg (2023)	1	3	1	3	2

Table 65: MRQ6 summary table for Chatterjee and Khovalyg (2023), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Chatterjee and Khovalyg (2023)	Strongly simulator- and benchmarking-aware, with moderate open-artifact critique	The survey’s dominant infrastructure concern is that RL-based HVAC research remains heavily simulation-dependent and lacks sufficiently rich, shared, open, and interoperable virtual infrastructure for fair cross-study validation and benchmarking.	The survey contributes strongly to MRQ6 by showing that comparability in RL-HVAC research depends on clear reporting of simulation environments, control settings, building types, and implementation status, while broader progress requires more standardized, open, and interoperable virtual testbeds rather than isolated simulation studies alone.

Table 66: One-line MRQ6 answer for Chatterjee and Khovalyg (2023), after recoding.

survey_id	MRQ6 answer
ChatterjeeKhovalyg2023	Chatterjee and Khovalyg (2023) answer MRQ6 by showing that RL-based HVAC research is still dominated by simulation environments, only partly validated in real buildings, and in need of a data-rich, open-source, interoperable virtual testbed to enable clearer cross-study validation, benchmarking, and more comparable infrastructure practices.

12 Al Sayed et al.

Table 67: MRQ6 recoded audit table for Al Sayed et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
AlSayedEtAl2024	3	3	3	The review is strongly environment- and data-realism-aware. Reproducibility auditing receives score 3 because the paper structurally audits training settings, simulator/model categories, exogenous-state sequences, and the generalization consequences of repeated or insufficiently diverse training conditions. Infrastructure auditing receives score 3 because the review gives detailed attention to training environments and simulation platforms, including EnergyPlus, TRNSYS, Dymola, BOPTEST, IESVE, CityLearn, MATLAB, and TESP, and distinguishes white-box, grey-box, and black-box approaches. Blind-spot visibility receives score 3 because weak diversification of exogenous state components such as occupancy schedules, electricity prices, and weather is treated as a central blind spot for reproducibility, generalization, and deployment.

Table 68: MRQ6 infrastructure-coverage scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Al Sayed et al. (2024)	3	3	2	0	3	3

Table 69: MRQ6 critical-audit scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Al Sayed et al. (2024)	2	0	3	3	3

Table 70: MRQ6 recurring infrastructure-gap scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Al Sayed et al. (2024)	0	2	3	3	3

Table 71: MRQ6 summary table for Al Sayed et al. (2024), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Al Sayed et al. (2024)	Strongly environment- and data-realism-aware, with moderate open-artifact critique	The survey’s dominant infrastructure concern is that RL-HVAC training environments often fail to represent exogenous variability realistically enough, especially weather, occupancy, and electricity-price variation, which weakens transfer from simulation to real deployment.	The survey contributes strongly to MRQ6 by showing that reproducibility and deployment readiness depend heavily on simulator choice, exogenously state design, and real-world data realism, even though the paper is less focused on open artifacts or code-sharing practices.

Table 72: One-line MRQ6 answer for Al Sayed et al. (2024), after recoding.

survey_id	MRQ6 answer
AlSayedEtAl2024	Al Sayed et al. (2024) answer MRQ6 mainly by showing that RL-for-HVAC research depends strongly on simulator realism, exogenous-state diversity, and real-world data support, and that weak handling of weather, occupancy, and price variability can undermine both reproducibility and later deployment.

13 Pinthurat et al.

Table 73: MRQ6 recoded audit table for Pinthurat et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
PinthuratEtAl2024	1	2	2	The review is broadly infrastructure-aware but only lightly reproducibility-critical. Reproducibility auditing receives score 1 because the paper does not systematically audit open code, benchmark environments, dataset sharing, artifact availability, or reproducible experimental packages. Infrastructure auditing receives score 2 because the paper frames RL-based smart home energy management with storage as depending on controllers, sensors, actuators, connectivity, software platforms, smart meters, grid interaction, renewable integration, energy storage, pricing signals, weather conditions, user preferences, demand-response signals, and grid state. Blind-spot visibility receives score 2 because the paper identifies sample inefficiency, data intensity, real-world testing needs, and sensitivity to actual user behavior, weather, and grid dynamics as practical limitations, although not as a strong open-benchmark or open-artifact agenda.

Table 74: MRQ6 infrastructure-coverage scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Pinthurat et al. (2024)	1	2	0	0	3	2

Table 75: MRQ6 critical-audit scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Pinthurat et al. (2024)	0	0	0	2	0

Table 76: MRQ6 recurring infrastructure-gap scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Pinthurat et al. (2024)	0	0	2	0	2

Table 77: MRQ6 summary table for Pinthurat et al. (2024), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Pinthurat et al. (2024)	Broadly infrastructure-aware, but only lightly reproducibility-critical	The survey’s dominant infrastructure concern is that RL-based smart-home energy management with storage depends heavily on external signals and broader smart-home system components, while real-world validation under changing user, environments, open artifacts, and weather, and grid conditions remains limited.	The survey contributes to MRQ6 mainly by making exogenous-signal dependence and smart-home system infrastructure explicit, but it remains weak on benchmark and stronger reproducibility critique.

Table 78: One-line MRQ6 answer for Pinthurat et al. (2024), after recoding.

survey_id	MRQ6 answer
PinthuratEtAl2024	Pinthurat et al. (2024) answer MRQ6 mainly by showing that RL-based smart-home energy management with storage depends on a broad control and data infrastructure and is especially sensitive to exogenous variables such as pricing, weather, user behavior, and grid dynamics, while giving only limited attention to shared benchmarks, open code, or artifact reproducibility.

14 Latoń et al.

Table 79: MRQ6 recoded audit table for Latoń et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
LatonEtAl2024	2	3	3	The review is strongly infrastructure-aware, with moderate reproducibility critique. Reproducibility auditing receives score 2 because the paper discusses data frameworks, privacy/security, interoperability, digital twins, and real-world case-study gaps, but gives little direct attention to open code or artifact availability. Infrastructure auditing receives score 3 because the paper structurally frames RL/DRL-based HEMS as depending on IoT devices, distributed control networks, smart-grid interaction, cloud and edge computing, data-collection pipelines, communication protocols, privacy-preserving mechanisms, digital twins, and scalable deployment infrastructure. Blind-spot visibility receives score 3 because limited real-world case studies, noisy and incomplete data, interoperability challenges, scalability limits, privacy/security risks, and insufficiently mature digital-twin infrastructure are explicit problems and future needs.

Table 80: MRQ6 infrastructure-coverage scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Latoń et al. (2024)	3	3	2	0	3	3

Table 81: MRQ6 critical-audit scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Latoń et al. (2024)	2	0	1	3	3

Table 82: MRQ6 recurring infrastructure-gap scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Latoń et al. (2024)	0	2	2	2	3

Table 83: MRQ6 summary table for Latoń et al. (2024), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Latoń et al. (2024)	Strongly infrastructure-aware, with moderate reproducibility critique	The survey’s dominant infrastructure concern is that RL/DRL-based HEMS depends on robust data pipelines, interoperable communication infrastructure, privacy and security mechanisms, scalable deployment infrastructure, and realistic digital-twin environments, while real-world case evidence remains limited.	The survey contributes strongly to MRQ6 by showing that reproducibility and practical comparability in HEMS are constrained not only by simulation-heavy evaluation, but also by weak data realism, interoperability challenges, scalability limits, privacy/security requirements, and insufficiently mature real-world infrastructure.

Table 84: One-line MRQ6 answer for Latoń et al. (2024), after recoding.

survey_id	MRQ6 answer
LatonEtAl2024	Latoń et al. (2024) answer MRQ6 mainly by showing that RL/DRL-based HEMS depends on substantial data, communication, privacy, interoperability, scalability, and digital-twin infrastructure, and that stronger benchmarking and real-world case evidence are still needed before the field can claim mature and broadly reusable research infrastructure.

Table 85: MRQ6 recoded audit table for Huynh et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
HuynhEtAl2025	3	3	3	The review is strongly data- and reproducibility-aware, with explicit attention to open-source practices. Reproducibility auditing receives score 3 because the paper explicitly includes open data and code-sharing practices as part of its contribution and records open-source implementations, programming language, simulation environment, historical-data use, weather and pricing inputs, and reproducibility-supporting tools such as OpenAI Gym and Docker. Infrastructure auditing receives score 3 because the paper makes simulation/testbed infrastructure, historical data, real-world input streams, programming platforms, and open-source implementation support structurally visible. Blind-spot visibility receives score 3 because data quality, noisy or incomplete sensor and communication data, computational complexity, heterogeneity across building types and HVAC configurations, scalability, interpretability, transparency, and stakeholder trust are recurring barriers to reusable and deployable MORL infrastructure.

Table 86: MRQ6 infrastructure-coverage scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Huynh et al. (2025)	3	3	2	3	3	3

Table 87: MRQ6 critical-audit scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Huynh et al. (2025)	2	3	2	3	3

Table 88: MRQ6 recurring infrastructure-gap scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Huynh et al. (2025)	2	2	2	2	3

Table 89: MRQ6 summary table for Huynh et al. (2025), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Huynh et al. (2025)	Strongly data- and reproducibility-aware, with explicit attention to open-source practices	The survey’s dominant infrastructure concern is that MORL for smart buildings depends on high-quality data, open and reusable implementation support, simulation/testbed infrastructure, and methods that can handle heterogeneous buildings, external signals, noisy real-world data streams, and computational scalability.	The survey contributes strongly to MRQ6 by making code-sharing practices, simulation/testbed support, real-world data inputs, and infrastructure heterogeneity visible, while also showing that stronger reproducibility still depends on better data quality, broader reuse, and more adaptable methods across diverse building settings.

Table 90: One-line MRQ6 answer for Huynh et al. (2025), after recoding.

survey_id	MRQ6 answer
HuynhEtAl2025	Huynh et al. (2025) answer MRQ6 by showing that MORL for smart buildings relies on substantial data, simulation/testbed, and open-source implementation infrastructure, while practical reproducibility and reuse are still constrained by noisy or incomplete data, heterogeneous building contexts, computational complexity, and limited standardization across studies.

Table 91: MRQ6 recoded audit table for Kannari et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
KannariEtAl2025	3	3	3	The review is strongly implementation- and validation-infrastructure aware, with a high level of reproducibility critique. Reproducibility auditing receives score 3 because the paper discusses comparable validation protocols, short and heterogeneous test periods, black-box market solutions, changing weather and occupancy, safety checks, fallback logic, governance, transparency, accountability, and the need for agreed digital-twin validation criteria. Infrastructure auditing receives score 3 because real-building RL is framed as depending on virtual pretraining, online deployment design, relearning loops, data preprocessing, recalibration, safety checks, digital-twin maturity, data sharing, security mechanisms, and IoT maturity. Blind-spot visibility receives score 3 because weak standardization of digital twins, limited data sharing, short test periods, black-box implementation practices, privacy risks, and nonstandard validation criteria are explicit limitations for reproducible real-building RL.

Table 92: MRQ6 infrastructure-coverage scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Kannari et al. (2025)	3	3	2	1	2	3

Table 93: MRQ6 critical-audit scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Kannari et al. (2025)	3	1	2	3	3

Table 94: MRQ6 recurring infrastructure-gap scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Kannari et al. (2025)	2	2	2	2	3

Table 95: MRQ6 summary table for Kannari et al. (2025), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Kannari et al. (2025)	Strongly implementation- and validation-infrastructure aware, with a high level of reproducibility critique	The survey’s dominant infrastructure concern is that real-building reinforcement learning still lacks sufficiently mature and standardized pretraining, deployment, validation, digital-twin, data-sharing, and governance infrastructure for reliable cross-case comparison and wider adoption.	The survey contributes strongly to MRQ6 by showing that reproducibility in real-building reinforcement learning depends not only on algorithms, but also on comparable validation protocols, data quality management, digital-twin maturity, security and privacy mechanisms, transparency, governance, and implementation practices.

Table 96: One-line MRQ6 answer for Kannari et al. (2025), after recoding.

survey_id	MRQ6 answer
KannariEtAl2025	Kannari et al. (2025) answer MRQ6 by showing that real-building reinforcement learning depends on robust pretraining, online deployment, validation, digital-twin, data-sharing, and governance infrastructure, and that reproducibility remains limited by short test periods, changing weather and occupancy conditions, nonstandardized validation criteria, limited transparency, and weak data-sharing practices.

17 Michailidis et al.

Table 97: MRQ6 recoded audit table for Michailidis et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
MichailidisEtAl2025	1	2	1	The review is structured and implementation-aware, but only lightly reproducibility-critical. Reproducibility auditing receives score 1 because the paper uses structured literature analysis and summary tables, but gives limited direct attention to open-source code, shared datasets, benchmark ecosystems, or artifact reproducibility. Infrastructure auditing receives score 2 because the paper makes implementation attributes and external-signal assumptions visible through dimensions such as RL methodology, agent framework, target outputs, baseline control strategies, RES configurations, BEMS types, building typologies, electricity prices, demand-response requests, weather, occupancy, renewable generation, and battery state of charge. Blind-spot visibility receives score 1 because some implementation and external-signal assumptions are visible, but reproducibility gaps are not a major organizing critique.

Table 98: MRQ6 infrastructure-coverage scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Michailidis et al. (2025)	2	1	0	0	3	3

Table 99: MRQ6 critical-audit scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Michailidis et al. (2025)	1	0	0	1	1

Table 100: MRQ6 recurring infrastructure-gap scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Michailidis et al. (2025)	0	0	1	0	1

Table 101: MRQ6 summary table for Michailidis et al. (2025), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Michailidis et al. (2025)	Structured and implementation-aware, but only lightly reproducibility-critical	The survey’s dominant infrastructure concern is that RL-based renewable-energy building control depends on clearly specified application context, tributes and external-signal assumptions visible through structured sum-baseline comparisons, RES configurations, BEMS types, building typologies, mary tables, but it remains weaker than the strongest MRQ6 papers on and exogenous signals such as prices, weather, occupancy, renewable gener- ation, and battery state of charge.	The survey contributes to MRQ6 mainly by making implementation at- tributes and external-signal assumptions visible through structured sum- open-source practices, benchmark infrastructure, and explicit reproducibil- ity critique.

Table 102: One-line MRQ6 answer for Michailidis et al. (2025), after recoding.

survey_id	MRQ6 answer
MichailidisEtAl2025	Michailidis et al. (2025) answer MRQ6 mainly by showing that RL-based renewable-energy building control is shaped by application context, baseline choice, BEMS and RES configuration, and strong dependence on external signals such as weather, pricing, occupancy, renewable generation, and battery state, while giving only limited attention to shared benchmarks, open code, or artifact reproducibility.

18 Li et al.

Table 103: MRQ6 recoded audit table for Li et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
LiEtAl2025	0	2	1	Reproducibility auditing receives score 0 because the survey contains no reproducibility vocabulary at all: the terms code, open-source, repository, benchmark and reproducibility do not occur anywhere in the paper, and the availability of artifacts from the reviewed studies is never raised. Infrastructure auditing receives score 2 because two subsections of the challenge framework treat infrastructure at paragraph level. Data collection and processing is discussed through dependence on data quality, sparse sensor deployment, rare failure data producing incomplete datasets, heterogeneous sources, solar intermittency and noisy high-dimensional data, with mitigations including virtual sensors, synthetic data generation and federated learning for privacy and scarcity. Environment construction and simulation is discussed through modeling complexity, computational demand and the discrepancy between simulated and real environments. Neither subsection names a specific simulator, dataset or platform, so no infrastructure landscape is surveyed. Blind-spot visibility receives score 1 because data scarcity and simulation fidelity are presented as technical obstacles facing individual studies rather than as shortcomings in the shared infrastructure or reporting practices of the literature, and no standardization need is identified.

Table 104: MRQ6 infrastructure-coverage scores for Li et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Li et al. (2025)	2	2	0	0	2	2

Table 105: MRQ6 critical-treatment scores for Li et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Li et al. (2025)	0	0	0	2	0

Table 106: MRQ6 gap-signal scores for Li et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Li et al. (2025)	0	0	0	0	2

Table 107: MRQ6 summary table for Li et al. (2025), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Li et al. (2025)	Absent, data-challenge oriented	The survey’s infrastructure concern is the quality and sufficiency of input data. Sparse sensors, rare failure records and heterogeneous sources are presented as the reason reinforcement learning underperforms in this domain, and the response offered is technical rather than collective: virtual sensors, synthetic data generation, federated learning and improved preprocessing. Simulation appears alongside this as an environment whose fidelity must be improved rather than as an artifact to be shared.	For MRQ6, the paper marks the floor of the corpus. It identifies a genuine data-realism problem and proposes remedies for it, but conducts no reproducibility audit of any kind, names no simulator, dataset or platform, and never raises code availability, benchmark comparability, simulator fragmentation or the need for shared infrastructure. The reviewed studies are described by what they achieved, never by what would be needed to repeat them.

Table 108: One-line MRQ6 answer for Li et al. (2025), after recoding.

survey_id	MRQ6 answer
LiEtAl2025	Li et al. (2025) treat infrastructure only as a data problem, discussing incomplete datasets from sparse sensors and rare failure events alongside simulation fidelity within a lifecycle challenge framework, while conducting no reproducibility audit whatsoever: code availability, shared benchmarks, simulator fragmentation and standardization are absent from the survey entirely, and no simulator, dataset or platform is named anywhere in the paper.

Table 109: MRQ6 recoded audit table for Lei et al. (2026). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
LeiEtAl2026	1	3	3	Reproducibility auditing receives score 1 because reproducibility enters only obliquely, through the call for standardized benchmarking protocols, while the availability of code or artifacts from the reviewed studies is never examined. The survey’s own reproducibility statement concerns its search procedure rather than the literature, and the remaining occurrences of the term describe digital twins failing to reproduce cascading failure dynamics. Infrastructure auditing receives score 3 because data infrastructure organizes a dedicated challenge subsection and recurs across the future-directions chapter. Extreme events are said to be rarely captured in historical building energy datasets, producing a severe imbalance between nominal and emergency operating conditions, and three remedies are assessed in turn with their residual limitations: digital twin environments augmented with physics-based disturbance models to generate synthetic trajectories, offline reinforcement learning drawing on operational records and fault logs, and reward shaping. The application table additionally records which studies used real-world datasets for calibration, and the future-directions chapter proposes policy repository management, cross-building experience sharing, and time-series foundation models pre-trained on large-scale multi-building datasets. Blind-spot visibility receives score 3 because the survey states plainly that no existing method fully resolves the tension between data availability and event rarity, that synthetic data may not reproduce cascading failure dynamics for event types outside the simulation model’s design envelope, that offline policies trained on normal-operation datasets underperform under distributional shift, and that heterogeneity in system scale, energy carrier composition, baseline choice and metric definition prevents direct cross-study comparison.

Table 110: MRQ6 infrastructure-coverage scores for Lei et al. (2026), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Lei et al. (2026)	2	3	1	0	2	3

Table 111: MRQ6 critical-treatment scores for Lei et al. (2026), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Lei et al. (2026)	3	0	1	3	3

Table 112: MRQ6 gap-signal scores for Lei et al. (2026), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Lei et al. (2026)	0	3	2	1	3

Table 113: MRQ6 summary table for Lei et al. (2026), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Lei et al. (2026)	Data-scarcity centered, standardization-seeking	The survey’s infrastructure concern is that the events it cares about are structurally absent from the data. Because extreme conditions are rare, historical records over-represent normal operation, so agents are trained where the problem is not and deployed where it is. Every proposed remedy is then assessed for whether it actually closes that gap, and each is found narrow sense: synthetic trajectories bounded by the simulation model’s design envelope, offline policies exposed to distributional shift, and shaped rewards at risk of optimizing auxiliary signals instead of load survival.	For MRQ6, the paper contributes an unusually disciplined data critique paired with an explicit call for shared infrastructure, including standardized benchmarking protocols, policy repository and cross-building experience sharing. What it does not do is audit reproducibility in the simulator platform is named, so the infrastructure argument is made about data and benchmarks rather than about the artifacts that would let a study be repeated.

Table 114: One-line MRQ6 answer for Lei et al. (2026), after recoding.

survey_id	MRQ6 answer
LeiEtAl2026	Lei et al. (2026) build a dedicated data-scarcity critique around the structural absence of extreme events from historical building datasets, assess digital twins, offline learning and reward shaping as remedies while naming what each fails to resolve, and call for standardized benchmarking protocols, policy repositories and cross-building experience sharing; code and artifact availability of the reviewed studies is never examined and no simulator platform is named.

Table 115: MRQ6 recoded audit table for Togashi (2025). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Togashi2025	1	1	3	Reproducibility auditing receives score 1 because the survey makes one reporting-quality observation about the literature, noting that in cases where the original designer’s intent was not fully explained the abstraction required conjecture, and acknowledging that its own simplified formulations would not necessarily produce identical optimization outcomes if applied to the original agents. Code and artifact availability of the reviewed studies is never raised, and the terms code, repository and open-source do not occur. Infrastructure auditing receives score 1 because the survey contains almost no infrastructure content: dataset, simulator, EnergyPlus, Modelica and CityLearn do not occur anywhere, and the only named resources are a testbed cited once for its key performance indicators and a comfort standard offered as a grounding for deadband boundaries. Blind-spot visibility receives score 3 because identifying an underexamined element of practice is the purpose of the review. Reward function design is presented as a relatively underexplored topic in existing review literature, sixty-eight unique designs are found among seventy-nine studies and named as severely impeding research comparability, weighting coefficients are found to rely on empirical derivation without clear theoretical basis, and no standard methodology is found for combining the four common design techniques.

Table 116: MRQ6 infrastructure-coverage scores for Togashi (2025), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Togashi (2025)	1	0	1	0	2	3

Table 117: MRQ6 critical-treatment scores for Togashi (2025), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Togashi (2025)	2	1	0	0	3

Table 118: MRQ6 gap-signal scores for Togashi (2025), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Togashi (2025)	0	1	2	0	1

Table 119: MRQ6 summary table for Togashi (2025), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Togashi (2025)	Notational standardization, computational infrastructure absent	The survey’s concern is that reward functions cannot be compared because they are not written in a common language. Its response is a standardization apparatus operating entirely on paper: unified variable symbols, integrated energy terms, a single signed quantity for electricity purchase and sale, a taxonomy of comfort indicators, normalization functions, and a standardized representation of weighting coefficients. Simulators, datasets, benchmarks and code play no part in the argument.	For MRQ6, the paper is an unusual case: it pursues standardization more determinedly than any other survey in the corpus while auditing infrastructure less than any of them. The distinction matters for how the corpus is read, because standardization here means notational convention rather than shared computational resources, and a survey can therefore score highly on the need for standardization while contributing nothing on simulators, datasets, benchmarks or artifact availability.

Table 120: One-line MRQ6 answer for Togashi (2025), after recoding.

survey_id	MRQ6 answer
Togashi2025	Togashi (2025) builds the review around a standardization methodology and states plainly that sixty-eight distinct reward designs across seventy-nine studies severely impede comparability, but the standardization is notational rather than infrastructural: the survey unifies symbols, units, weighting representations and comfort indicator taxonomies while never mentioning datasets, simulators, benchmarks or code availability anywhere in the paper.

Table 121: MRQ6 recoded audit table for Mazhar et al. (2026). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
MazharEtAl2026	1	3	2	Reproducibility auditing receives score 1 because open-source tooling is named as a resource practitioners can adopt, including a distributed reinforcement learning framework, a distributed training library and a cloud platform, but the availability of code or artifacts from the reviewed studies is never examined. The term reproducibility occurs once and does not concern the literature, and the terms repository and documentation do not occur. Infrastructure auditing receives score 3 because a dedicated subsection catalogues the datasets and benchmarking environments available for validation, naming four datasets with their characteristics, including one covering over six hundred buildings and one supplying high-resolution occupancy-based demand data, alongside five simulation and benchmarking environments, and because a separate subsection surveys the frameworks used for model development and scalable training. Blind-spot visibility receives score 2 because the survey identifies the lack of real-world benchmarks among its stated research gaps, states that high variability in data formats and standards complicates integration, and calls for large-scale real-time data from multi-zone buildings, but frames these as shortages of available infrastructure rather than as reporting or transparency failures in the reviewed literature.

Table 122: MRQ6 infrastructure-coverage scores for Mazhar et al. (2026), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Mazhar et al. (2026)	3	3	2	2	2	2

Table 123: MRQ6 critical-treatment scores for Mazhar et al. (2026), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Mazhar et al. (2026)	1	0	0	2	3

Table 124: MRQ6 gap-signal scores for Mazhar et al. (2026), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Mazhar et al. (2026)	0	2	2	1	2

Table 125: MRQ6 summary table for Mazhar et al. (2026), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Mazhar et al. (2026)	Infrastructure-cataloguing, standards-oriented	The survey’s concern is that the components needed to build and connect systems are fragmented rather than missing. Datasets, simulation environments and training frameworks are inventoried as available resources, while the obstacle identified is that devices from different manufacturers use proprietary protocols and inconsistent data formats. The response is standardization: open messaging protocols, middleware unifying disparate systems, ontology-based device descriptions, edge gateways for protocol translation, and an eventual phase of industry-wide standard setting.	For MRQ6, the paper supplies the corpus’s most complete inventory of available infrastructure while conducting no reproducibility audit of the work it reviews. Naming datasets, environments and frameworks tells the reader what exists; it does not establish what any reviewed study used, released or documented. The standardization argument is correspondingly about device and data interoperability for future deployments rather than about making published results repeatable.

Table 126: One-line MRQ6 answer for Mazhar et al. (2026), after recoding.

survey_id	MRQ6 answer
MazharEtAl2026	Mazhar et al. (2026) catalogue infrastructure more thoroughly than any other survey in the corpus, naming four datasets, five simulation and benchmarking environments and several training frameworks, and argue at length for interoperability standards covering messaging protocols, middleware, ontology-based device descriptions and edge protocol translation; the reviewed studies themselves are never audited for code or artifact availability, so the argument concerns what exists for future work rather than what published results rest on.

Table 127: MRQ6 recoded audit table for Mohsenpour and Xing (2026). Scores use the shared 0–3 evidence-intensity scale.

survey__id	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
MohsenpourXing2026	1	3	2	Reproducibility auditing receives score 1 because the survey makes one transparency observation, noting for two of the catalogued instruments that accuracy specifications are not disclosed, while the availability of code or artifacts from the reviewed studies is never examined and the single occurrence of the term reproducibility describes the review’s own screening process. Infrastructure auditing receives score 3 because tooling is inventoried at three levels and tied to the reviewed literature. A simulation framework figure names the building simulation platforms, the reinforcement learning environment libraries, and the co-simulation middleware that couples them for online learning; a summary table records which simulation platforms the reviewed studies actually implemented; and a dedicated data-sources subsection with its own table catalogues the environmental, physiological and building-system sensors used, listing measurement ranges and stated accuracies for each. Blind-spot visibility receives score 2 because the survey observes that simulation environments do not capture the complexities and uncertainties of operational buildings and records where instrument accuracy is undocumented, but it identifies no shortage of shared benchmarks, no fragmentation problem across the many platforms it lists, and no need for standardization.

Table 128: MRQ6 infrastructure-coverage scores for Mohsenpour and Xing (2026), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Mohsenpour and Xing (2026)	3	3	1	2	2	3

Table 129: MRQ6 critical-treatment scores for Mohsenpour and Xing (2026), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Mohsenpour and Xing (2026)	0	1	1	2	0

Table 130: MRQ6 gap-signal scores for Mohsenpour and Xing (2026), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Mohsenpour and Xing (2026)	0	0	1	1	2

Table 131: MRQ6 summary table for Mohsenpour and Xing (2026), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main MRQ6 consequence
Mohsenpour and Xing (2026)	Tool-inventorying, standardization absent	The survey’s concern is instrumentation: what a researcher would need in order to build and sense the system. It traces the pipeline from building simulation platforms through reinforcement learning libraries and co-simulation middleware to the physical sensors that supply environmental, physiological and building-system measurements, giving model numbers, measurement ranges and accuracies. Shared conventions between studies are not part of the argument.	For MRQ6, the paper supplies the corpus’s most granular account of what the literature is built from, descending to instrument accuracy in degrees Celsius and parts per million, while making no collective claim about that infrastructure. It lists many platforms without treating the variety as fragmentation, records which studies used which simulator without asking whether results are therefore comparable, and never raises code availability, shared benchmarks or standardization. Detail substitutes for critique.

Table 132: One-line MRQ6 answer for Mohsenpour and Xing (2026), after recoding.

survey__id	MRQ6 answer
MohsenpourXing2026	Mohsenpour and Xing (2026) document infrastructure in more granular detail than any other survey in the corpus, naming the building simulation platforms, reinforcement learning libraries and co-simulation middleware in a pipeline figure, recording which platforms the reviewed studies implemented, and cataloguing each sensor with its measurement range and stated accuracy; the survey nonetheless draws no collective conclusion from this, raising neither code availability nor shared benchmarks nor standardization anywhere in the paper.